

Benchtop Instrument

TGem Pro Spectrophotometer

New generation of intelligent user-friendly nucleic acid detector



Manual

Inquire



Catalog Number | Packaging

Mat. No	Ref. No	No. of p
OSE-260-03	OSE-260-03	1 se
OSE-260-05	OSE-260-05	1 set
OSE-260-06	OSE-260-06	1 set

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Description

TGem Pro spectrophotometer is a fixed wavelength ultra-micro ultraviolet-visible spectrophotometer (hereinafter referred to as TGem Pro). It not only provides a base mode (as shown in Figure 1-1) for measuring low dose samples, but also can use a traditional cuvette (as shown in Figure 1-2) for sample testing.

In base mode, this product can measure 0.5-2 μl samples with extremely high accuracy and repeatability. When the base mode is used to measure samples, the sample retention system applies surface tension to keep the samples between the two measuring optical fibers. In this way, the samples can be tested with higher concentration without dilution. The measurement range can reach 200 times of the concentration range tested by ordinary spectrophotometer, and the liquid column height between optical fibers can be adjusted in real time to obtain more accurate measure data. Compared with the traditional cuvette testing method, the base mode eliminates the complex cuvette cleaning process. The dust-free paper is convenient to clean. In addition, for samples with low concentration, high volatility or low surface tension that are not easy to form a liquid column, a cuvette can be used for measurement.

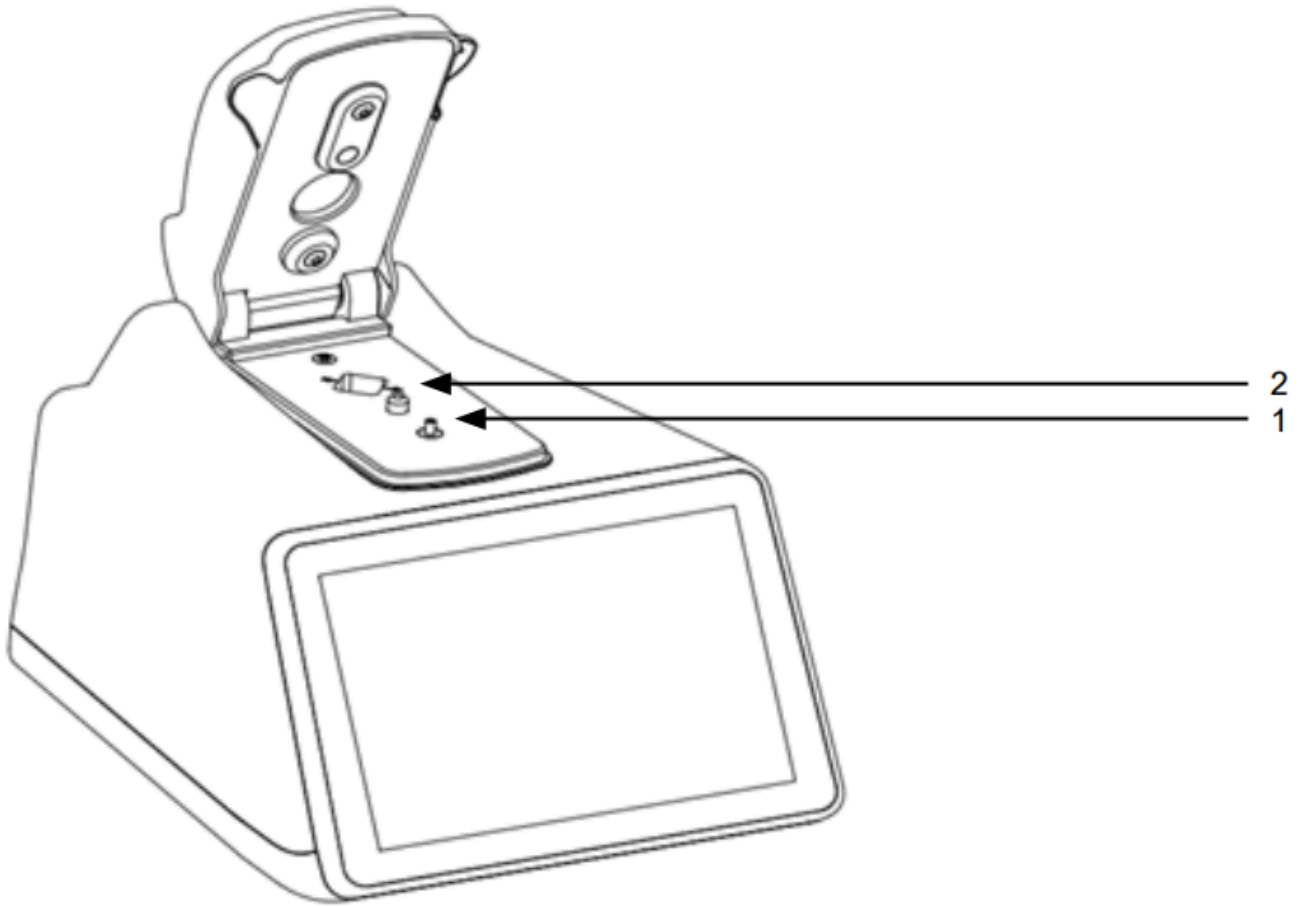


Figure 1

Parameters



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 Cat. No.		Home Product OEM/ODM Resources About TIANGEN	OSE-260-03	 English
Testing unit		3648 Pixel linear CCD array		
Minimum sample amount (μl)		0.5		
Path length (mm)		Automatic switching between 0.03/0.05/0.1/0.2/1.0		
Wavelength range (nm)		230/260/280/340/600		
Wavelength accuracy (nm)		±1		
Spectral resolution (nm)		≤1.8(FWHM at Hg 253.7 nm)		
Absorbance accuracy		0.002Abs (1 mm optical path), or 1% CV, and take the larger value		
Absorbance accuracy		±2%(at 0.64A at 350 nm)		
Minimum measure concentration of base		2 ng/μl dsDNA		
Maximum measure concentration of base		27500 ng/μl dsDNA		
Testing time		≤5 seconds		
Cuvette	Heating temperature (°C)	37±0.5		
	Stir speed (RPM)	10-900 rpm 10 adjustable gears		
	Absorbance range	0-1.5A (10 mm)		
	Minimum measure concentration	0.2 ng/μl dsDNA		
Display screen		7-inch, 1280*800 pixel color display screen, capacitive touch screen, support gesture operation		
Operating system		Android		
Internal storage space		8 GB		
Data interface		USB, WIFI		
Power supply		AC110V-220V, 50Hz/60Hz(Power adapter)		
Weight		2.3 KG		



Package Contents

Product name	Quantity	Unit	Inquire 
Host	1	Set	
Power cord	1	Piece	
Power Adapter	1	Piece	
User manual	1	Piece	
Product measure report	1	Piece	
Conformity certificate	1	Piece	
Product warranty card	1	Piece	

Precautions

Before turning on the power switch of the instrument, open the upper arm to take away the anti-collision gasket, and then lower the upper arm gently, as shown in Figure 2. This anti-collision gasket is to prevent the upper arm from colliding with the positioning column during the movement of the instrument. After use of the instrument, it is recommended to put the anti-collision gasket back to the position shown in Figure 2.

Before using the instrument, connect the 12V adapter power cord to the power interface on the back of the instrument, insert the three-core plug into the power socket, and turn on the instrument power switch to start the instrument, as shown in Figure 3; Both USB type A interface can be used for product program upgrade and data export.

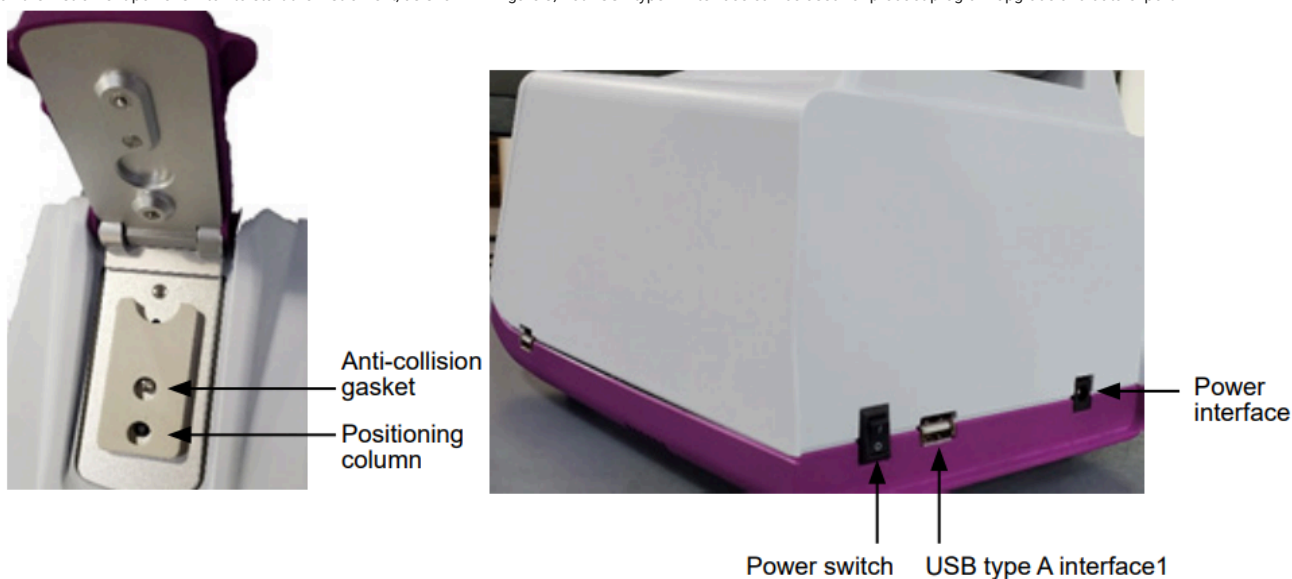


Figure 2

Figure 3

When the base mode and cuvette mode are switched, a prompt of channel switching progress will pop up on the screen. Do not open the upper arm during the channel switching process, and wait for the channel switching to succeed before proceeding to the next step, as shown in Figure 4.

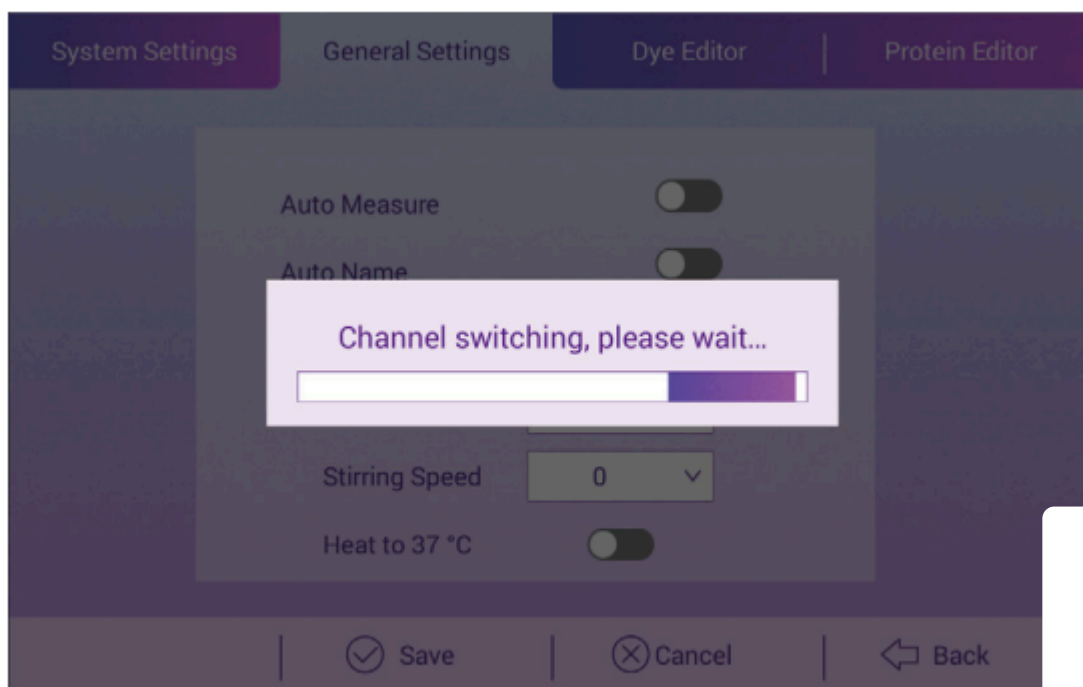


Figure 4

Quality Assurance

The warranty period of TGem Pro spectrophotometer is 12 months. If the instrument is damaged due to improper operation, unqualified use and unauthorized maintenance or modification by users, it will not be covered by the warranty. For instrument maintenance after the warranty period, TIANGEN Company will charge maintenance fees according to after-sales service standards.

Precautions	- After the instrument is unpacked, the items in the packing box shall be accepted immediately according to the packing list. In case of damaged or missing items, please contact the supplier immediately. - After unpacking the instrument, please keep the packing boxes and packaging materials properly for maintenance. TIANGEN BIOTECH (BEIJING) CO., LTD will not be responsible for the instrument damage caused by improper packaging on the way to the maintenance department.
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Important Notes

• Custom and usage:

 Precautions	The items contain particularly important information. Please read it carefully. If you fail to follow the instructions, it may lead to malfunction or even instrument damage.
 Warning	The warning message requires you to be very careful in operating a certain step or method. If you do not follow the requirements correctly, it may lead to serious personal injury.

• Safety:

In the whole process of operation, maintenance and repair of this instrument, the following basic safety measures and warnings and precautions pointed out in this user's manual must be observed. If these measures or the warnings and precautions pointed out in this manual are not observed, the basic protection provided by the instrument may be affected. At the same time, it will not meet the safety standards of instrument design and manufacture, and affect the expected application range of the instrument. TIANGEN BIOTECH (BEIJING) CO., LTD shall not bear any responsibility for all consequences caused by the user's failure to comply with the following requirements.

Precautions	The instrument is Class I equipment conforming to GB4793.1 standard, with protection grade of IP20. This instrument is a product for indoor use.
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• Instrument grounding

In order to avoid electric shock, the input power cord of the instrument must be grounded reliably. The instrument adopts the safety device, that is, the three-core grounding plug with a third (grounding) pin, which can only be used with a grounding power socket. If the plug cannot be inserted into the socket, please ask the electrician to install the correct socket and prevent the grounding plug from losing its safety protection function.

• Keep away from live circuits

The operator shall not disassemble the instrument protection, replace the components or adjust the instrument without authorization. If necessary, it must be certified professional maintenance personnel. It is strictly prohibited to replace the components when the power supply is connected.

• Pay attention to power supply

Before switching on the AC power supply, ensure that the voltage of the power supply is consistent with the voltage required by the instrument (12 V) and ensure load of the power socket is not less than the maximum load of the instrument of 30 W.

• Pay attention to the power cord

The instrument should use the supplied power cord. If the power cord is broken, it must be replaced. Do not repair it by yourself. It shall be replaced with a pov type and specification. When the instrument is in use, do not place anything on the power cord and do not place the power cord in a place where people flow.

• Power cord plugging

When plugging and unplugging the power cord, please hold the operating part of the plug correctly. When inserting the plug, ensure that the plug is completely into the socket. Do not pull the power cord when unplugging.

• Pay attention to the placement of instruments

The instrument shall be placed in a place without corrosive gas and smoke, and there shall be no strong light, strong air flow or strong magnetic field interference in the room. The work table on which the instrument is placed shall be level and stable.

Turn off the power supply when stopping working. If it is not in use for a long time, cut off the power supply, unplug the power plug, and cover the instrument with soft cloth or plastic film to prevent dust and foreign matter from entering.

• Operation precautions

During the measure operation, avoid liquid dripping on the instrument;

When it is tested in base mode, the sample shall be added to the corresponding position of the instrument;

When it is tested in the cuvette mode, the cuvette shall be placed in the corresponding position of the instrument.



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Precautions

Under the following circumstances, immediately cut off the power supply, unplug the instrument from the power socket, and contact the supplier or ask qualified maintenance personnel to handle it:

- There is liquid falling into the instrument;
- The instrument gets wet due to rain or water;
- The instrument fails to work normally, especially with any abnormal noise or smell;
- The instrument is fallen down or the shell is damaged;
- The function of the instrument has changed obviously.

- Operation and maintenance of instrument:
- Base maintenance

Sample compatibility statement:
The base part of the TGen Pro is accessible to most solvents and solutions commonly used in life science laboratories, including methanol, ethanol, propanol, isopropanol, butanol, acetone, ether, chloroform, carbon tetrachloride, DMSO, DMF, acetonitrile, tetrahydrofuran, toluene, hexane, benzene, sodium hydroxide, sodium hypochlorite, dilute hydrochloric acid, dilute nitric acid and dilute acetic acid.
Base operation and maintenance:
Please wipe the liquid from the base immediately after testing the sample. If testing is not continued, please clean the base with pure water and dry it. In this way, the solution or solvent will not cause damage to the base.
Do not contact with any form of hydrogen fluoride (HF), and fluoride ions will dissolve the silica fiber in the base.
Do not allow any liquid to enter the gap between the base and the body, otherwise the instrument may be damaged. If liquid spills, wipe it immediately.
• Operation and maintenance of the complete machine
Please read the User's Manual carefully before using the product.
Avoid direct sunlight. Avoid direct wind blowing during operation, so as not to affect the accuracy.
Avoid using in the environment with high humidity as far as possible.
Keep surfaces clean by wiping them frequently with a clean soft cloth. Strictly prohibit the corrosive cleaning agents.
If there is any abnormal situation, please record the process and phenomenon in detail, and contact our company in time.



The power supply must be cut off while the instrument is being cleaned.
The surface of the instrument is strictly forbidden to be cleaned with corrosive cleaning agents.

Sort by

	Product	Resources	About TIANGEN	OEM/ODM
Building 5, No. 86, Shuangying West Road, Changping District, Beijing. +86 10 59822688 people@tiangen.com	Genomic DNA	Manual Download	TIANGEN Introduction	× Inquire ^
	RNA	Literature	Applications & Public Events	
	Plasmid DNA & DNA Clean Up	Technical Resources	News	
	PCR	Visual Operation Guide	Contact Us	
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	Next-Generation Sequencing			
	Epigenetics			
	Instrument			
	Cloning Reagent			
	Marker & Electrophoresis Reagent			
	Sample Preparation			
	Raw Materials			